N-of-1 Trials and Precision Medicine: A Detailed Lexicon

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Purpose. A working glossary of the terminology used across the N-of-1 / single-case trial and precision-medicine literature. Each entry pairs a technical term with a plain-language definition pitched at an undergraduate statistics major, and lists the alternative names the same concept carries in different fields (biostatistics, psychometrics / single-case research, and precision / personalized medicine).

Provenance. Terms were mined from the full text of N-of-1 and precision-medicine articles in the project Zotero library (`nof1` collection, 72 curated articles; full text obtained and machine-read for 42). The statistical and psychometric supplement was compiled from standard methodology references. See the closing *Provenance and status* section for scope and caveats.

How to read the synonym column. Where biostatistics, psychometrics, and precision medicine use different names for the *same* underlying concept, all variants are listed together so the same idea can be recognized whatever vocabulary a given paper adopts. A condensed same-concept / different-name map appears in Section 12.

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Notation index and glossary

The lexicon is written in plain language and uses few symbols. Where an entry names one, it maps onto `analysis/report/NOTATION.md` as follows; symbols marked *local* have a meaning specific to a single entry and are not the canonical project symbol.

Table 1: Notation index and glossary of symbols used in this document

Symbol (as written in an entry)	Meaning	Status
ρ (`rho`)	AR(1) serial correlation; also used for the ICC in some sources	canonical (AR(1)); local (ICC)
α	significance level, the Type I error rate	canonical
β (beta), $1 - \beta$	Type II error rate; power, written π in project papers	local; canonical as π
λ (`lambda`)	carryover decay rate in the exponential-decay entry	canonical
λ (smoothing parameter)	spline roughness penalty	local
τ^2 (tau-squared)	between-patient (between-study) variance in meta-analysis	local
Σ, Ω	covariance matrices	local
X, Z	fixed-effects and random-effects design matrices	local
t	time	canonical

Sections 1 to 13 are themselves the glossary. Where a British spelling appears in a synonym column (for example ‘personalised’), it is listed deliberately as a variant that readers will meet in the literature.

1. Trial designs and architectures

Table 2: Trial design and architecture terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
N-of-1 trial	A controlled experiment run inside a single patient, who is switched between treatments multiple times so the best treatment *for that one person* can be estimated.	single-patient trial; single-subject trial; single-case experimental design (SCED, psychology); individualized / idiographic clinical trial; one-person trial; n=1 RCT
Aggregated N-of-1 trial	A set of separate single-patient trials pooled together to estimate both a population-average effect and sharpened individual effects.	series / combined / pooled N-of-1 trials; aggregated personalized studies; multiple N-of-1 study
Single-case experimental design (SCED)	The behavioral-science family of designs that study one subject repeatedly across baseline and treatment phases; N-of-1 trials are the randomized-crossover member of this family.	single-subject design; single-case design; single-subject research
Crossover trial	A trial in which each participant receives the treatments in sequence over separate periods, serving as their own control.	cross-over / change-over trial; within-subject / within-patient design; repeated-measures comparison
AB/BA (2x2) crossover	The simplest crossover: half the subjects get A then B, half get B then A, in two periods.	two-period two-treatment design; classical crossover
Multiple-crossover design	A crossover in which the patient switches back and forth between treatments several times rather than once.	multi-crossover; repeated-period crossover; double crossover

Term	Plain-language definition	Synonyms / aliases (by field)
Parallel-group trial	A trial in which each participant receives only one treatment and group averages are compared between people.	parallel-group RCT (PG-RCT); between-participant / between-patient design
Randomized controlled trial (RCT)	A study that assigns participants to treatments by chance to give an unbiased comparison.	controlled trial; randomized trial
AB / ABAB design	Single-case notation for alternating a baseline/treatment (A) and an alternate treatment (B); ABAB introduces, withdraws, and reintroduces treatment.	reversal design; withdrawal design; challenge-withdrawal design
Multiple-baseline design	A single-case design that staggers when treatment starts across people, settings, or behaviors to rule out coincidental change.	staggered-baseline design
Changing-criterion design	A single-case design in which the success threshold is shifted in steps and the outcome is shown to track each new target.	–
Alternating-treatments design	A single-case design that rapidly switches among two or more treatments within one subject to compare them.	multi-element design
Open-label (OL) design	A study or phase in which both patient and staff know which (active) treatment is given.	unblinded design

Term	Plain-language definition	Synonyms / aliases (by field)
Blinded discontinuation (BDC)	A phase in which active treatment is secretly replaced by placebo to see whether benefit fades.	open-label + blinded discontinuation (OL+BDC)
Micro-randomized trial (MRT)	A within-person trial (often via phone or wearable) that randomizes a brief intervention many times to build just-in-time adaptive interventions.	micro-randomised trial (British spelling)
Interrupted time-series design	Repeated measurements on one unit compared in level/slope before and after a defined intervention point.	ITS; single-case time series
Adaptive design	A trial allowing prospectively planned changes (e.g., dropping arms, reallocation) based on accumulating data.	response-adaptive / sequential design; group-sequential design
Play-the-winner / response-adaptive randomization	An adaptive scheme that shifts the randomization odds toward the treatment performing better.	response-adaptive randomization
Biomarker-stratified trial	A trial that splits randomization by a baseline biomarker and tests the treatment within each biomarker group and overall.	biomarker-driven / stratified design
Enrichment / adaptive-enrichment design	A trial that preferentially enrolls (or, after an interim look, narrows to) the patients most likely to respond.	population enrichment; adaptive enrichment

Term	Plain-language definition	Synonyms / aliases (by field)
Basket trial	One treatment tested across multiple diseases that share a common molecular target.	–
Umbrella trial	Multiple treatments matched to different biomarkers within a single disease.	–
Platform trial	A standing trial framework evaluating multiple treatments against a shared control, adding or dropping arms over time.	–
Two-stage / multi-stage design	A trial run in phases with an interim analysis between them that can change later conduct.	two-stage adaptive design

2. Periods, sequence, carryover, and washout

Table 3: Period, sequence, carryover, and washout terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
Treatment period	A block of time during which a patient stays on one assigned treatment before switching.	period; treatment block; phase; intervention administration period
Cycle	One full pass through the treatments being compared (e.g., one A-then-B pair).	pair; treatment pair; block
Sequence	The full ordered list of treatment periods a participant goes through.	treatment sequence; allocation sequence

Term	Plain-language definition	Synonyms / aliases (by field)
Washout period	A treatment-free gap inserted so leftover effects of the prior treatment disappear before the next is assessed.	wash-out; clearance / rest interval
Active (analytical) washout	Switching directly to the next treatment but not *measuring* until the previous drug's effect has gone, used when stopping treatment would harm the patient.	analytical washout
Run-in / titration phase	A preliminary phase before randomization used to stabilize dose, confirm eligibility, clear prior drugs, or assess adherence.	lead-in; stabilization phase; dose titration / escalation
Baseline phase	The pre-treatment stretch of repeated measurements used as the comparison reference.	pre-intervention phase; phase A
Carryover effect	When a treatment's effect persists into and contaminates the following period.	carry-over; residual / lingering effect
Simple carryover	A carryover whose size is the same regardless of which treatment follows; assumed to last only one period.	first-order carry-over
Complex carryover	A carryover whose size or direction depends on the specific order or pairing of treatments.	treatment-sequence-dependent carryover

Term	Plain-language definition	Synonyms / aliases (by field)
Behavioral / psychometric carryover	A non-pharmacological carryover where earlier treatment changes habits, adherence, or how a patient remembers rating themselves, mimicking a drug carryover.	behavioral carry-over; recall carryover
Period effect	A systematic change in outcome tied to which time period it is, independent of treatment.	time / secular / nominal trend; cycle effect
Order / sequence effect	A bias arising because the order in which treatments are given affects the response.	treatment-ordering effect
Treatment-by-period interaction	When the treatment difference is not the same across periods, often a sign of carryover or instability.	period-by-treatment interaction
Counterbalancing	Building a balanced (non-random) treatment order such as ABBA to offset known time trends.	counterbalanced design
Condition- vs time-dependent crossover rule	Whether to switch treatments once the patient returns to baseline (condition-dependent) or after a fixed wait (time-dependent).	crossover rule
On/off kinetics	How quickly a drug starts working and wears off, which sets how long a washout must be.	rapid onset / offset; drug half-life

3. Randomization, blinding, and conduct

Table 4: Randomization, blinding, and conduct terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
Randomization	Assigning treatment or treatment order by chance to remove systematic bias.	random allocation / assignment
Block (permuted-block) randomization	Randomizing within small balanced groups so each treatment appears equally often and bad runs are avoided.	blocking; restricted randomization; counterbalancing
Stratified randomization	Randomizing separately within subgroups (site, sex) to keep those factors balanced.	stratification; minimization; covariate-adaptive allocation
Allocation concealment	Hiding the upcoming assignment from those enrolling participants so it cannot be foreseen or manipulated.	concealed allocation
Blinding	Concealing which treatment is given from patients and/or assessors to prevent expectation bias.	masking; single / double / triple blind
Placebo / placebo effect	An inert comparator made to look identical to the active treatment; the placebo effect is improvement wrongly credited to that inert treatment.	sham / dummy treatment; expectancy response
Active comparator / control	The reference treatment (placebo, usual care, or an established drug) against which the test treatment is judged.	comparator; control condition; reference arm

Equipoise	Genuine uncertainty about which compared treatment is better, which ethically justifies randomizing.	clinical equipoise

4. Analysis models

Table 5: Analysis model terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
Linear mixed-effects model	A regression combining population-level fixed effects with subject-specific random effects, suited to repeated measurements per person.	mixed / mixed-effects model; multilevel model (MLM); hierarchical linear model (HLM); random-effects model; LME; MMRM (mixed model for repeated measures)
Multilevel / hierarchical model	A model with parameters nested at several levels (measurements within patients within studies) that share information across levels.	nested model; HLM; intensive hierarchical model
Fixed effect	A model term whose coefficient is a single constant assumed to apply to everyone.	population-average / marginal effect
Random effect	A model term whose value varies across subjects, treated as a draw from a distribution.	subject-specific effect; random coefficient
Random intercept	A per-subject baseline offset letting each individual start at their own level.	subject-specific intercept

Term	Plain-language definition	Synonyms / aliases (by field)
Random slope	A per-subject deviation in how strongly the outcome changes with a predictor (e.g., over time).	subject-specific slope; random trend
Design matrix	The table of predictor values mapping coefficients to observations; separate ones for the fixed (X) and random (Z) parts.	model matrix; X / Z matrix
Marginal model	A model describing the *population-average* outcome rather than a specific individual's.	population-average model
Conditional model	A model describing the outcome for a specific individual given their own random effects.	subject-specific model
Generalized estimating equations (GEE)	A method for correlated/repeated outcomes that estimates the population-average effect using a chosen working correlation, staying valid even if that correlation is mis-specified.	GEE; Zeger-Liang / marginal approach; PGEE (penalized)
Working correlation structure	The assumed (possibly approximate) within-subject correlation pattern GEE uses.	independence / exchangeable / autoregressive working structure
Generalized linear (mixed) model	A regression for non-normal outcomes (counts, proportions, binary), optionally with random effects.	GLM; GLMM
Ordinary least squares (OLS)	Fitting a regression by minimizing the sum of squared residuals.	least-squares regression

Term	Plain-language definition	Synonyms / aliases (by field)
Generalized least squares (GLS)	Regression that accounts for correlated or unequal-variance errors, giving the best linear unbiased estimate.	weighted least squares (related); BLUE estimation
Analysis of covariance (ANCOVA)	A regression that adjusts the outcome for a baseline covariate to improve precision.	baseline adjustment
ANHECOVA	Covariate adjustment that includes all treatment-by-covariate interactions; asymptotically it is never less efficient than the unadjusted or ANCOVA estimator.	analysis of heterogeneous covariance
Change score	The outcome minus its baseline value, a simple (often inefficient) way to adjust for baseline.	difference score; baseline subtraction
Proportional hazards (Cox) model	A survival regression in which covariates multiply a baseline rate of an event over time.	Cox regression; coxph
Proportional odds model	An ordinal-outcome regression assuming covariate effects are constant across category thresholds.	cumulative logit model
Dynamic regression model	A regression that explains the present outcome using lagged (past) values of itself and of predictors, absorbing autocorrelation.	autoregressive distributed-lag model; dynamic modeling
Paired t-test	A test comparing two related measurements (e.g., one patient on A vs B) by analyzing their differences.	matched-pairs / Student's t-test

Term	Plain-language definition	Synonyms / aliases (by field)
Serial t-test	A t-test for one person's repeated data that corrects the variance and degrees of freedom for serial correlation.	–
Randomization (permutation) test	A significance test that compares the observed effect with effects from many random reassignments of the treatment schedule.	permutation test; randomization-test reference set
Visual analysis	Judging a single-case treatment effect by inspecting the graphed time series (level, trend, variability) rather than by formal statistics.	visual inspection; graphical analysis
Nonoverlap index	A single-case effect size based on how little the treatment-phase data overlap the baseline data.	overlap statistic (see Tau-U, NAP, PND in Section 9)
Meta-analysis	Statistically pooling results across studies (or N-of-1 trials) into one overall estimate.	research / quantitative synthesis; mega-analysis
Individual participant data (IPD) meta-analysis	Pooling the raw per-patient data across studies, usually via mixed models, rather than just their summaries.	one-stage meta-analysis
Network meta-analysis	Combining trials of different treatment pairs to compare treatments never tested head-to-head.	mixed-treatment / indirect comparison
Intention-to-treat (ITT)	Analyzing participants by their assigned treatment regardless of adherence, preserving randomization.	ITT (modified ITT excludes those with no outcome data)

Term	Plain-language definition	Synonyms / aliases (by field)
Per-protocol analysis	Analyzing only participants who followed the protocol, which can introduce bias.	completers / per-protocol population

5. Serial correlation and time series

Table 6: Serial correlation and time-series terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
Serial correlation	The tendency for measurements taken close together in time on the same person to be related rather than independent.	autocorrelation; serial dependence; within-subject / intra-subject correlation
Autocorrelation function (ACF)	A function or plot giving the correlation of a series with itself at each successive lag.	–
Lag	The number of time steps separating two observations being compared.	–
First-order autoregressive process, AR(1)	A model where each measurement correlates with the previous one and correlation decays as `rho` raised to the time gap.	AR(1); lag-1 autocorrelation; corAR1 / corCAR1 (continuous-time)
ARMA / ARIMA	Time-series models combining autoregressive and moving-average (and, for ARIMA, differencing) terms.	autoregressive integrated moving average

Term	Plain-language definition	Synonyms / aliases (by field)
Stationarity	The assumption that a series' mean, variance, and correlation structure stay constant over time.	stationary time series
Detrending	Removing a systematic upward/downward trend so fluctuations can be studied.	–
Pre-whitening	Removing autocorrelation from a series before analysis, which can accidentally remove a real effect.	whitening
Cochrane-Orcutt / Prais-Winsten	Older regression procedures that adjust estimates for first-order autocorrelated errors.	–
Effective sample size	The smaller number of *independent* observations that correlated data are statistically worth; shrinks under positive correlation.	effective number of independent observations
Slope-autocorrelation indeterminacy	The difficulty, in short series, of telling apart a genuine trend from trend produced by autocorrelated errors.	confounding of slope and autocorrelation

6. Variance, power, and estimation

Table 7: Variance, power, and estimation terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
Components of variation	The separate sources of variability in an outcome (between-treatment, between-patient, patient-by-treatment, within-patient).	variance components; sources of variation
Between-patient variance	Variability in average outcomes from one patient to another.	between-subject variance; tau-squared (meta-analysis)
Within-patient variance	Variability across repeated occasions in the same patient on the same treatment.	within-subject / residual / error variance
Intraclass correlation coefficient (ICC)	The fraction of total variability due to differences between subjects; equivalently, how alike a subject's repeated measures are.	ICC; rho; intra-subject correlation
Covariance / variance-covariance matrix	A table of the variances and pairwise correlations of repeated measures or parameter estimates.	Sigma / Omega matrix; dispersion matrix
Compound symmetry (CS)	A covariance pattern assuming every pair of repeated measures is equally correlated.	exchangeable correlation
Unstructured covariance	A covariance pattern estimating every variance and correlation freely with no imposed shape.	general covariance
Toeplitz covariance	A covariance pattern where the correlation depends only on how many timepoints apart two measures are.	banded covariance

Term	Plain-language definition	Synonyms / aliases (by field)
Kronecker correlation	Building a large correlation matrix as the structured product of smaller ones (e.g., within-period x between-period).	Kronecker-product structure
Positive definite	A required property of a valid covariance matrix: every weighted combination of the variables has positive variance.	PD
Heteroscedasticity	Unequal variance of the outcome across conditions or over time.	heteroskedasticity; unequal variances
Variogram	A plot of how the variance of differences between measurements grows with the time gap, used to detect autocorrelation.	semivariogram
Maximum likelihood estimation (MLE)	Choosing parameter values that make the observed data most probable.	ML; likelihood-based estimation
Restricted maximum likelihood (REML)	A maximum-likelihood variant that estimates variance components with less bias by adjusting for the fixed effects.	restricted / residual maximum likelihood
Likelihood ratio test (LRT)	A test comparing two nested models by the difference in their fit.	deviance chi-squared test
Wald test	A test judging whether a coefficient differs from zero using its estimate divided by its standard error.	Wald statistic

Term	Plain-language definition	Synonyms / aliases (by field)
Kenward-Roger approximation	A small-sample degrees-of-freedom correction improving fixed-effect tests in mixed models.	KR correction
Empirical Bayes / BLUP	A subject-specific estimate that borrows strength from the group mean, shrinking noisy individual estimates toward it.	best linear unbiased predictor; shrinkage estimate
Shrinkage	Pulling individual or extreme estimates toward a common value to reduce noise and overfitting.	regularization; penalization; regression to the mean; James-Stein shrinkage
Borrowing strength	Using data from other individuals or studies to sharpen a sparse unit's estimate.	partial pooling; shrinkage
Penalized splines / B-splines	Flexible curve-fitting building blocks with a roughness penalty, used to model smooth nonlinear effects without overfitting.	B-spline basis; smoothing / roughness penalty
Smoothing parameter (lambda)	A tuning knob controlling how wiggly versus smooth a fitted curve is.	penalization / tuning parameter
Non-centrality parameter	A quantity in a test statistic's distribution under the alternative hypothesis; larger values mean higher power.	NCP
Type I error	Falsely declaring an effect that does not exist (a false positive); its rate is the significance level alpha.	alpha; size of the test; false-positive rate

Term	Plain-language definition	Synonyms / aliases (by field)
Type II error	Failing to detect a real effect (a false negative); its complement is power.	beta; false negative
Statistical power	The probability of correctly detecting a true effect of a given size.	1 - beta; sensitivity of a test
Sample-size / power calculation	Computing how many patients (or periods/measurements) are needed for adequate power or precision.	power analysis; study planning
Effect size	A standardized measure of how large a treatment difference is, often in standard-deviation units.	standardized mean difference (SMD); Cohen's d; tau
Bias (of an estimate)	A systematic tendency for an estimate to be too high or too low on average.	systematic error
Mean squared error (MSE)	An accuracy summary combining an estimator's variance and squared bias.	RMSE (its square root); bias-variance trade-off
Confidence interval	A range that, over many repeats, would contain the true value a stated percentage of the time.	CI; interval estimate
Coverage	How often computed intervals actually contain the true value across repeated studies/simulations.	interval coverage
Monte Carlo simulation	Repeatedly generating random datasets with known truth to study how a method or design behaves.	simulation study

Term	Plain-language definition	Synonyms / aliases (by field)
Multivariate normal distribution	A bell-shaped joint distribution for several correlated variables, set by a mean vector and covariance matrix, used to generate repeated-measures data.	MVN; joint normal
Bootstrap	Resampling the observed data (with replacement) to approximate the sampling distribution of a statistic.	parametric / wild / multiplier bootstrap
Pitman asymptotic relative efficiency	A ratio comparing how many subjects two designs need to reach the same power.	relative efficiency
Information criterion (AIC / BIC / QIC)	Scores ranking models by fit while penalizing extra parameters; lower is better (QIC is the GEE analogue of AIC).	Akaike / Bayesian / quasi-likelihood information criterion

7. Bayesian methods

Table 8: Bayesian methods terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
Bayesian inference	Updating prior beliefs with data via Bayes' rule to obtain a posterior probability distribution over parameters.	Bayesian analysis
Prior distribution	The probability distribution encoding belief about a parameter before seeing the data.	prior

Term	Plain-language definition	Synonyms / aliases (by field)
Posterior distribution	The updated distribution for a parameter after combining prior and data.	posterior
Posterior predictive distribution	The distribution of new/future data implied by the fitted model, used for prediction and model checking.	predictive distribution
Noninformative prior	A deliberately vague prior that lets the data dominate.	flat / diffuse / weak / reference prior
Informative prior	A prior carrying real external knowledge or expert opinion.	–
Weakly informative prior	A prior that gently constrains estimates to plausible ranges without strongly steering results.	–
Conjugate prior	A prior whose form yields a posterior in the same family, simplifying computation.	(e.g., inverse-gamma prior for a variance)
Credible interval	A Bayesian interval containing the parameter with a stated probability given the data.	posterior interval; Bayesian confidence interval
Posterior probability of benefit	The Bayesian probability that a treatment's true effect is favorable, read off the posterior.	–
Hierarchical / partial pooling	Estimating group-specific values as a compromise between each group's own data and the overall average.	partial pooling; Bayesian shrinkage

Term	Plain-language definition	Synonyms / aliases (by field)
Exchangeability	Treating individuals' effects as interchangeable draws from a common distribution.	–
Markov chain Monte Carlo (MCMC)	A simulation algorithm that draws samples from a complex posterior to enable Bayesian computation.	MCMC; Gibbs / Metropolis sampling; JAGS / WinBUGS / Stan estimation
Posterior predictive check	A model check comparing data simulated from the fitted model against the observed data.	Bayesian p-value
Prior elicitation	Formally turning experts' beliefs into a prior distribution, especially influential when data are scarce.	expert-opinion elicitation
Prior effective sample size	A way to express how much information a prior contributes by equating it to a number of observations.	prior ESS; ECSS / EHSS (effective current / historical sample size)
Prior-data conflict	Disagreement between the prior and the observed data, which can distort inference unless the prior discounts it.	prior-likelihood conflict; commensurability
Robust / mixture / power / commensurate prior	Priors designed to down-weight historical or external information when it conflicts with current data.	meta-analytic-predictive (MAP) prior; discounting prior
Sequential probability ratio test (SPRT)	A method that tests a hypothesis after each new observation and stops once evidence is conclusive.	SPRT; bSPRT (bootstrapped); sequential analysis

8. Missing data and dropout

Table 9: Missing data and dropout terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
Dropout / attrition	Participants leaving the study before completion, leaving incomplete data.	withdrawal; loss to follow-up; discontinuation; noncompletion
Differential dropout	Unequal (or systematically different) dropout between treatment arms.	differential / relative attrition
Biased / informative dropout	When those who leave differ systematically (e.g., responders quitting when switched to placebo), distorting results.	informative dropout
Missing completely at random (MCAR)	Missingness unrelated to any observed or unobserved data.	MCAR
Missing at random (MAR)	Missingness explainable by observed data, so it can be validly modeled / imputed.	ignorable missingness
Missing not at random (MNAR)	Missingness that depends on the unseen value itself and cannot be fully corrected from observed data.	nonignorable / informative missingness (IM)
Last observation carried forward (LOCF)	Filling a missing value with the patient's last recorded value; simple but often biased.	–
Multiple imputation (MI)	Filling missing values several times from their predicted distribution and combining results to reflect uncertainty.	imputation
Complete-case analysis	Analyzing only subjects with no missing data, which can bias results.	listwise deletion; completers analysis

Term	Plain-language definition	Synonyms / aliases (by field)
Inverse probability weighting (IPW)	Reweighting observed cases to stand in for similar dropouts.	inverse-probability weights
Selection model	A missing-data model that predicts the chance of dropping out from the (possibly unseen) outcome.	–
Pattern-mixture model	A missing-data model letting completers and dropouts have different outcome distributions.	–
Sensitivity analysis	Re-running an analysis under alternative assumptions to see how robust the conclusions are.	robustness analysis

9. Single-case / psychometric effect measures

Table 10: Single-case and psychometric effect measures with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
Level change	An abrupt shift in the average outcome from one phase to the next.	mean shift; phase jump; intercept shift
Trend / rate change	A change in the slope (rate of improvement or decline) between phases.	slope change; time-by-phase interaction; drift
Tau-U	A nonoverlap effect size for single-case data that quantifies improvement, optionally correcting for baseline trend.	–
Nonoverlap of all pairs (NAP)	The proportion of treatment-phase points exceeding baseline points across all pairwise comparisons.	–

Term	Plain-language definition	Synonyms / aliases (by field)
Percentage of nonoverlapping data (PND)	The percentage of treatment points beyond the most extreme baseline point.	–
Percentage of all nonoverlapping data (PAND)	The proportion of data that would need removing to eliminate overlap between phases.	–
Standardized mean difference (SMD)	The mean difference between phases or treatments divided by a standard deviation, allowing comparison across scales.	between-case SMD; standardized effect size
Multilevel meta-analysis of single cases	Combining many single-case datasets in a hierarchical model to estimate average effects and their variation across cases.	MultiSCED-type analysis
Simulation Modeling Analysis (SMA)	A bootstrap-like resampling method generating null replicates to get empirical p-values for single-case data.	–
Piecewise / segmented regression	Fitting separate linear segments to different phases (baseline vs treatment) of a series.	spline / discontinuity growth model
Idiographic	Focused on patterns within a single individual.	within-person approach
Nomothetic	Focused on general laws or averages across many individuals.	population approach
Ecological momentary assessment (EMA)	Repeatedly sampling a person's experiences in real time in their natural environment.	experience sampling

Term	Plain-language definition	Synonyms / aliases (by field)
Reliability	The degree to which a measurement yields consistent, reproducible results.	(test-retest, internal-consistency reliability)
Standard error of measurement	The expected spread of measurement errors around a person's true score.	–
Measurement error	The discrepancy between an observed measurement and the true underlying value.	observation error; noise
Floor / ceiling effect	Scores clustering at the lowest or highest possible value, hiding true differences.	–

10. Biomarkers, precision medicine, and treatment-effect heterogeneity

Table 11: Biomarker, precision-medicine, and heterogeneity terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
Precision medicine	Tailoring treatment to an individual's biological, genetic, physiological, and behavioral profile.	personalized / personalised (British spelling) / individualized / stratified / tailored medicine
Patient-by-treatment interaction	Genuine variation among patients in how strongly they respond to a treatment; the core target of personalized medicine.	treatment-effect heterogeneity (HTE); subject-by-treatment interaction; differential / individual response; effect modification
Heterogeneity of treatment effects (HTE)	The fact that the same treatment helps different patients by different amounts.	treatment-effect heterogeneity; differential treatment effect

Term	Plain-language definition	Synonyms / aliases (by field)
Effect modifier	A patient characteristic that changes the size or direction of a treatment's effect.	moderator (psychology); interaction variable; predictive marker
Biomarker-treatment interaction	The statistical effect showing a treatment's benefit depends on a patient's biomarker value; the key signal these designs test.	treatment-by-biomarker interaction; predictive interaction; effect modification
Predictive biomarker	A measurable patient trait that forecasts who will benefit (differentially) from a specific treatment and guides its selection.	predictive marker / signature / covariate; treatment-effect modifier; theranostic biomarker
Prognostic biomarker	A trait that predicts the likely course of disease regardless of treatment.	prognostic factor / variable / covariate; baseline covariate
Qualitative (biologic) interaction	An interaction where treatment helps some patients but harms others (the effect reverses direction); cannot be removed by rescaling.	crossover / non-removable interaction
Quantitative (statistical) interaction	An interaction in degree only (benefit varies in size but keeps direction); often removable by transforming the scale.	removable / scale-dependent interaction
Additivity	When the treatment effect is constant across patients on a chosen scale, so there is no interaction.	additive model
Individual treatment effect (ITE)	The difference in outcome a specific patient would have under treatment versus control.	individual-level effect

Term	Plain-language definition	Synonyms / aliases (by field)
Conditional average treatment effect (CATE)	The average treatment effect within a subgroup defined by particular characteristics.	–
Average treatment effect (ATE)	The mean treatment effect across the whole population.	population / main effect
Subgroup analysis	Examining treatment effects within predefined patient subsets; prone to spurious findings if unplanned.	subgroup comparison
Risk modeling (predictive HTE)	Stratifying patients by their predicted baseline risk to see who benefits most.	risk-stratified / risk-based subgrouping
Effect modeling (predictive HTE)	Building a model with treatment-by-covariate interactions to predict individual benefit.	interaction-based modeling
Responder / non-responder	A patient classified as benefiting or not, often by a threshold.	response classification
Biomarker-positive / -negative subgroup (B+ / B-)	The patients whose biomarker indicates they are more (B+) or less (B-) likely to benefit.	targeted / non-targeted subgroup
Biomarker prevalence	The fraction of the population that is biomarker-positive.	biomarker frequency
Companion diagnostic	A laboratory test required to identify patients eligible for a specific targeted therapy.	–
Potential outcomes / counterfactual	The pair of outcomes a unit would show under each treatment (only one is observed); the counterfactual is the unobserved one.	Neyman-Rubin model; potential outcome

Term	Plain-language definition	Synonyms / aliases (by field)
SUTVA	The assumption that one unit's treatment does not affect another's outcome and there is only one version of each treatment.	stable unit treatment value assumption
Pharmacogenomics	Using a patient's genetics to predict and tailor treatment response.	individualized / precision therapy
Learning health system	An infrastructure that continually uses routine clinical data to learn which patients benefit from which treatments.	rapid-learning health system

11. Clinical, pharmacology, and outcome terms

Table 12: Clinical, pharmacology, and outcome terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
Pharmacokinetic / pharmacodynamic (PK/PD) model	Models of how a drug's concentration changes over time (PK) and how concentration produces an effect (PD).	PK/PD model; Emax / dose-response model
Half-life	The time for a drug's concentration (or effect) to fall by half; governs carryover length.	t-half; elimination rate; clearance
Cumulative dose effect	A drug effect that depends on the whole dosing history, weighted toward recent doses.	dose-history dependence

Term	Plain-language definition	Synonyms / aliases (by field)
Modified Gompertz function	A three-parameter S-shaped curve (maximum response, rate, displacement) modeling a response rising to a plateau.	three-parameter Gompertz
Exponential decay	A model where an effect shrinks by half over a fixed time, used for carryover persistence (`exp(-lambda * t)`).	half-life / lambda decay
Regression to the mean	The tendency of extreme initial values to drift toward average on re-measurement, mimicking improvement.	–
Bioequivalence	Showing two drug formulations deliver the active ingredient similarly enough to be interchangeable.	average / population / individual bioequivalence
Prescribability / switchability	Whether a generic can be given to new patients (prescribability) or substituted in patients already on the brand (switchability).	population vs individual equivalence
Minimal clinically important difference (MCID)	The smallest change in an outcome that patients perceive as meaningful.	clinical-importance cut-off
Clinical vs statistical significance	Whether an effect is large enough to matter to a patient versus merely unlikely to be chance.	clinical significance / importance
Patient-reported outcome measure (PROM)	An outcome reported directly by the patient (symptoms, quality of life).	PRO; self-report outcome

Term	Plain-language definition	Synonyms / aliases (by field)
Surrogate outcome	An early or proxy measurement standing in for the true outcome of interest.	surrogate endpoint
Time-to-event endpoint	An outcome measured as the time until an event occurs, analyzed with survival methods.	survival endpoint; PFS / OS
Hazard rate / hazard ratio	The instantaneous risk of an event over time; the ratio comparing that risk between groups.	HR; baseline hazard
Absolute risk difference	The plain difference in outcome probability between treated and control patients.	risk difference; absolute treatment effect
Relative risk / risk ratio	The ratio of event probability in one group to another.	RR; relative risk reduction (its complement)
Odds ratio	A measure comparing the odds of an outcome between two groups.	OR
Censoring	When the event of interest is not observed within follow-up (e.g., the study ends first).	administrative / right censoring
Comparative effectiveness research (CER)	Research comparing real-world treatments head-to-head to see which works better for whom.	head-to-head comparison; patient-centered outcomes research (PCOR)
Evidence-based medicine (EBM)	Practicing medicine by applying the best current research evidence to patient care.	–
Therapeutic window	The dose range that is effective without causing harmful side effects.	therapeutic range

Term	Plain-language definition	Synonyms / aliases (by field)
Rare / ultra-rare disease	A condition affecting so few people that conventional group trials are infeasible, favoring N-of-1 designs.	low-volume treatment context
Actigraphy	Using a wrist-worn motion sensor to objectively estimate sleep and activity.	actigraph; accelerometry
Prazosin	An alpha-1 adrenoceptor antagonist studied for PTSD trauma nightmares and sleep disturbance (the project's running clinical example).	alpha-1 blocker
PTSD / CAPS	Post-traumatic stress disorder; the Clinician-Administered PTSD Scale (CAPS) is a structured interview rating its symptom frequency and intensity.	post-traumatic stress disorder; CAPS-IV

12. Cross-field synonym map (same concept, different name)

The single concept on the left is named differently depending on whether a paper is written from a biostatistics, single-case / psychometric, or precision-medicine perspective.

Table 13: Cross-field synonym map across biostatistics, psychometric, and precision-medicine usage

Underlying concept	Biostatistics term	Single-case / psychometric term	Precision-medicine term
One-patient repeated-crossover study	N-of-1 trial; single-patient trial	single-case experimental design (SCED); single-subject design	idiographic / individualized clinical trial
Genuine person-to-person variation in response	patient-by-treatment interaction; variance component	subject-by-treatment interaction	heterogeneity of treatment effects (HTE)
A variable that changes a treatment's effect	treatment-by-covariate interaction; effect modification	moderator	predictive biomarker / effect modifier
Subject-specific deviations in a model	random effects; variance components	between-case variance	(individual effects)
Pulling individual estimates toward the mean	shrinkage; empirical Bayes / BLUP	–	borrowing strength / partial pooling
Correlation of nearby-in-time measures	serial correlation; AR(1); ICC	autocorrelation	–
Combining many small studies/patients	mixed-model / IPD meta-analysis	multilevel meta-analysis of single cases	aggregated N-of-1 analysis
The unobserved alternative outcome	counterfactual; potential outcome	–	individual treatment effect (ITE)
Multiple measurements per unit over time	repeated measures; longitudinal data	single-case time series	–
Standardized magnitude of an effect	standardized mean difference; Cohen's d	Tau-U / NAP / PND (nonoverlap indices)	–

Underlying concept	Biostatistics term	Single-case / psychometric term	Precision-medicine term
Real-time repeated outcome capture	–	ecological momentary assessment (EMA)	micro-randomized trial sampling

13. Reporting guidelines and ethics

Table 14: Reporting guideline and ethics terms with definitions and synonyms

Term	Plain-language definition	Synonyms / aliases (by field)
CONSORT	The standard checklist for transparently reporting randomized controlled trials.	Consolidated Standards of Reporting Trials
CENT 2015	The CONSORT extension giving reporting standards specifically for N-of-1 trials.	CONSORT Extension for N-of-1 Trials
SPENT / SPIRIT	SPIRIT is the trial-protocol reporting guideline; SPENT is its N-of-1 protocol extension.	SPIRIT extension for N-of-1
RoBiNT scale	A risk-of-bias / methodological-quality rating scale for single-case and N-of-1 studies.	Risk of Bias in N-of-1 Trials scale
PATH statement	The Predictive Approaches to Treatment effect Heterogeneity reporting guidance for HTE analyses.	–

Term	Plain-language definition	Synonyms / aliases (by field)
TRIPOD / PROGRESS / ICEMAN	Reporting/appraisal frameworks for prediction models (TRIPOD), prognosis research (PROGRESS), and credibility of subgroup/interaction claims (ICEMAN).	–
EQUATOR Network	An organization promoting accurate, transparent reporting of health research via guidelines.	Enhancing Quality and Transparency Of health Research
Operating characteristics	A design's statistical performance profile (type I error, power, bias) across scenarios.	frequentist properties
Pre-specification (a priori analysis)	Defining hypotheses and analyses before seeing data, to avoid data-driven false findings.	pre-registration
Selective reporting / data dredging	Cherry-picking favorable analyses or subgroups, inflating false positives.	multiplicity; subgroup-fishing
Family-wise error rate (FWER)	The probability of at least one false positive across multiple tests.	multiplicity control
Alpha allocation / spending	Dividing the overall significance level across hypotheses or interim looks to control multiplicity.	alpha splitting / spending; gatekeeping; fixed-sequence testing
Internal vs external validity	Whether conclusions are valid within the studied sample (internal) versus generalize beyond it (external).	generalizability; applicability; representativeness

Term	Plain-language definition	Synonyms / aliases (by field)
Equipoise	Genuine uncertainty about which treatment is better, ethically justifying randomization.	clinical equipoise
Informed consent / IRB approval	Ethical safeguards requiring participant agreement and institutional ethics-board review before research.	ethics-committee approval; written consent
Patient and public involvement (PPI)	Involving patients and the public in designing, running, and interpreting research.	patient representatives

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Provenance and status

- **Corpus.** The `nof1` collection of the project Zotero library holds 72 curated N-of-1 / precision-medicine articles. Local full text was obtained for 46; 42 were successfully machine-read and mined for terms (4 were skipped owing to a content-filter false positive on one processing batch, but their concepts are redundantly covered by other articles). Three of the read files held mismatched or banner-only text (an unrelated BMJ news page, an arbovirus review, and a publisher download banner) and contributed no terms; one file was mislabeled in the catalog (a Bayesian effective-sample-size paper) and was mined on its actual content.
- **Definitions** were extracted by reading the source texts; plain-language wording is the lexicon authors' paraphrase, *verified* against the mined passages. The statistical / psychometric supplement (Sections 5-9 in part) draws on standard methodology references (Pinheiro & Bates; Fitzmaurice, Laird & Ware; Box, Jenkins & Reinsel; Gelman et al.; Imbens & Rubin; de Vet et al.; CENT; Senn) and is *inferred from established domain knowledge* rather than verified against a specific retrieved source.
- **Synonym groupings** reflect usage observed across the corpus plus conventional cross-field equivalences; they are aids to recognition, not claims of exact mathematical identity (e.g., a quantitative interaction and an additive-scale effect coincide only on a particular scale).